ArogyaNet Technical Report: The MediScan Audit

Aryan Pandey · Arogya Diagnostics ML audit · Repository: github.com/ltd-ARYAN-pvt/ArogyaNet · Status as of 9 November 2026: Part 1 complete, selected Part 2 items started

1. Summary

Rohit's slide claimed "ResNet50, 98.4% accuracy, ready for deployment". The pilot showed the model flagging 7 of 10 healthy children and missing two sick ones. This report audits the data behind that claim, rebuilds the model under honest evaluation rules, and tests the three instructions the CEO issued.

Main findings so far:

- **The data has integrity problems.** 30 groups of exact duplicate files exist inside the train or test splits, and perceptual hashing flags 26 train-test pairs as possible near-duplicates still awaiting visual verification. The original validation set holds only 16 images.
- **A shortcut warning.** A CNN that sees only a 16×16 version of each X-ray beats the majority baseline by about 15 to 17 points of accuracy, so coarse, non-anatomical signals carry label information.
- **Classical features are already strong.** HOG + logistic regression reaches 97.41% validation accuracy, matching a fine-tuned MobileNetV3Small (97.38% ± 0.31%). Deep learning has barely cleared the classical bar.
- **Our own network.** ArogyaNet's standard-CNN variant (A0, 1.94 M parameters) was the best performer at 97.77% ± 0.40%. The compact depthwise variants lost accuracy, and the hypothesis that metadata helps was not supported.
- **The CEO's second instruction is the wrong one.** Chasing 99% accuracy is the wrong target. At camp prevalence, the cost-weighted error at a chosen operating point is what matters (Section 9).

All validation numbers below are **internal** (random stratified splits of one hospital's data). None of them is evidence of clinical performance.

Progress against the assignment

Item	Status
Part 1, tasks 1–6 (audit, duplicates, validation, shortcut, baselines, transfer)	Done
Part 1, task 7 (ArogyaNet and ablations)	Done; ablation table in Section 6
Part 1, task 8 (two-branch experiment)	Done
Part 1, task 9 (resolution and augmentation ablations)	Not yet reported
Part 2, task 1 (cost matrix and threshold)	Done on validation data
Calibration and prior-shift correction	Done (extra analysis)
Part 2, tasks 2–8 (5-fold CV, tuning, ensembles, feature space, external hospital, deployment)	Pending

2. Setup and reproducibility

- **Framework:** TensorFlow/Keras, with scikit-learn, OpenCV, Pillow and scikit-image. Environment managed with `uv` (`pyproject.toml` and `uv.lock` pin all versions), Python version fixed in `.python-version`.
- **Seeds:** 42, 123 and 2026 for every multi-seed experiment.
- **Scripts:** each result comes from a script under `src/` (for example `src/audit/dataset_inventory.py`, `src/models/shortcut_cnn.py`, `src/models/hog_baseline.py`, `src/models/lbp_baseline.py`, `src/models/transfer_learning.py`). Figures are in `figures/part1/`, written reports in `reports/part1/`, and experiment logs under `experiments/`.
- **Data:** the Kaggle Chest X-Ray Images (Pneumonia) archive. Image files are excluded from the repository because of their size.
- **Validation protocol:** a stratified 80:20 train/validation split of the original training pool, with the original test split held back for final evaluation. **For the three-seed experiments each seed uses a different split**, so the reported standard deviations include split variation as well as training randomness. In the ArogyaNet ablations (Section 6) all five variants share the same split within a given seed.

3. Data audit

3.1 Inventory

The locally extracted dataset has 5,856 images, against 5,863 in the dataset description.

Split	NORMAL	PNEUMONIA	Total
Train	1,341	3,875	5,216
Test	234	390	624
Validation (as shipped)	8	8	16
Total	**1,583**	**4,273**	**5,856**

The training pool is about 74% pneumonia, against roughly 10% expected at a rural camp. This mismatch is why accuracy on this data is misleading and why the threshold work in Section 8 is needed.

The audit script also records image mode (true greyscale vs RGB), file sizes and per-class pixel statistics. Any systematic class difference in brightness or image size is a candidate shortcut, and Section 3.4 shows that coarse image properties alone carry label information.

3.2 Duplicates and leakage

Check	Result
Exact duplicates (SHA-256)	30 groups, 34 duplicate pairs
Exact duplicates across train, test and validation	None. All groups sit inside a single split
Near-duplicates across train and test (perceptual hash)	26 candidate pairs, awaiting visual verification

The exact duplicates inflate whichever split contains them but do not leak across splits. The 26 train-test candidates are the important finding. Each one, if confirmed, lets the test set reward memorisation. Until the pairs are inspected, the claim is only that overlap is **possible**, not proven. The worst five pairs are shown side by side in the figure below.

A second, unresolved risk is that the dataset has no patient identifiers. Several images may come from the same child, so even a random stratified split cannot guarantee patient-level independence. This is the main reason the internal numbers above should

not be read as clinical accuracy.

3.3 The 16-image validation set

The validation folder holds 8 NORMAL and 8 PNEUMONIA images. A perfect score on n = 16 has an exact (Clopper-Pearson) 95% confidence interval of **about 79.4% to 100%**, so it cannot distinguish a good model from a mediocre one. It is also far too small to tune thresholds or hyperparameters on.

The replacement design is a **stratified 80:20 train/validation split of the original training pool**, with the original 624-image test set left untouched for final evaluation. Stratified k-fold was kept for the later cross-validation work (Part 2, task 2) because it costs five training runs per model. For the many Part 1 comparisons a single stratified split is the cheaper option, and the fixed test set protects the final number.

3.4 Shortcut hunt: 16×16 inputs

A deliberately crippled CNN was trained on X-rays downsampled to 16×16 pixels. At that size no lung texture, rib edge or lesion boundary can be resolved. The repository contains two reported runs:

Run	Accuracy	Balanced accuracy	PR-AUC	NORMAL recall	PNEUMONIA recall
Run A	89.08%	80.69%	0.988	63.43% (98 false positives of 268)	97.94% (16 false negatives of 776)
Run B	91.76%	85.54%	n/a	n/a	n/a
Majority-class baseline	74.33%	50.00%	n/a	0%	100%

Both runs beat the majority baseline by 15 to 17 percentage points. **Interpretation:** substantial class information survives extreme downsampling, so the label correlates with coarse properties such as overall brightness, contrast, framing or acquisition style. A full-resolution model can use the same properties. The result is a **diagnostic warning, not proof** of shortcut learning, because the experiment does not say *which* property is responsible. A masked-lung or intensity-matched control would be needed to isolate it.

4. Classical baselines

Both baselines use 128×128 greyscale inputs and logistic regression on standardised features.

Model	Accuracy	Balanced acc.	Precision	Recall	F1
Majority class	74.33%	50.00%	n/a	100%	n/a
LBP (radius 1, 8 points, uniform; 10-dim histogram)	87.36%	84.04%	92.04%	90.85%	91.44%
HOG (8,100-dim)	97.41%	96.79%	98.45%	98.07%	98.26%

HOG's confusion matrix on the validation subset is 256 true negatives, 12 false positives, 15 false negatives and 761 true positives. LBP's global histogram keeps texture frequencies but discards where they occur, while HOG keeps spatially organised gradient orientation. That difference is a plausible but untested explanation of the gap. The comparison applies only to these two configurations.

The key point is that a simple classical pipeline already scores 97.41%. Any deep model has to beat that bar by a clear margin to justify its cost, and as the next section shows, MobileNetV3Small does not.

5. Transfer baseline

MobileNetV3Small (ImageNet weights) was fine-tuned in two stages: classification head only with the backbone frozen, then the last 30 backbone layers unfrozen with BatchNorm layers kept frozen. Greyscale images were replicated across three channels. The head is global average pooling, dropout and a sigmoid output. Results are mean ± SD over seeds 42, 123 and 2026.

Metric	Mean ± SD
Accuracy	97.38% ± 0.31%
Balanced accuracy	96.08% ± 0.08%
Precision	97.75% ± 0.27%
Recall	98.75% ± 0.71%
F1	98.25% ± 0.22%
PR-AUC	99.81% ± 0.05%

Pretrained low-level filters (edges, gradients, textures) are expected to transfer to radiographs even though ImageNet contains colour photographs, and the high score is consistent with that. However, the fine-tuned network matches the HOG baseline

(97.41% vs 97.38%), so the gain from ImageNet pre-training over hand-crafted gradients is effectively zero on this split. Because each seed used a different validation split, the SDs mix split and training variation.

6. ArogyaNet: design and ablations

6.1 Hypotheses (stated before results)

ID	Design decision	Hypothesis	Experiment
H1	Depthwise separable convolutions	Reduce complexity without excessive performance loss	Standard vs depthwise separable
H2	Residual connections	Improve optimisation and feature learning	Residual vs non-residual blocks
H3	Squeeze-and-excitation attention	Improve channel-wise feature selection	With vs without SE
H4	Global average pooling	Reduce classifier parameters	GAP vs flatten-and-dense

6.2 Variants and results

Five variants were trained on three seeds (42, 123, 2026), with an identical stratified 80:20 split for all variants within each seed.

Variant	Description	Parameters	Mean validation accuracy	PR-AUC
A0	Standard CNN	1,941,665	**97.77% ± 0.40%**	99.93% ± 0.01%
A1	Depthwise separable convolutions	232,001	91.89% ± 6.69%	n/a
A2	A1 + residual connections	n/a	96.58% ± 1.08%	n/a
A3	A2 + SE attention	n/a	similar to A2	slightly higher than A2
A4	A3 + multi-scale branch	n/a	no gain over A3	n/a

All variants are under the 2 M parameter limit. A0 is the strongest at 1.94 M parameters.

6.3 Honest reading of the hypotheses

- **H1 partly supported.** Depthwise separable convolutions cut parameters by about 88% (1,941,665 to 232,001), but mean accuracy fell to 91.89% with a very large spread (SD 6.69 points). That variation across seeds means the compact network trained unreliably, not just worse.
- **H2 supported.** Adding residual connections recovered accuracy to 96.58% and reduced the spread to 1.08 points, consistent with easier optimisation.
- **H3 inconclusive.** SE attention kept accuracy similar and gave a small PR-AUC improvement. That difference is within the seed noise and cannot be claimed as a real effect.
- **H4 not yet reported.** The GAP vs flatten ablation result is not in the current notes.
- The multi-scale variant (A4) did not improve on A3, so that design decision did not pay off.

A0 is kept as the performance benchmark and A3 as the compact candidate. The central lesson is a trade-off: the smallest networks cost accuracy and stability, and the most accurate model is also the largest.

7. Two-branch experiment

A3′s convolutional encoder was combined with three handcrafted numbers (mean intensity, aspect ratio, log image area) through a small dense branch, concatenated before the classifier, over three seeds with matched splits.

Model	Mean accuracy	Mean PR-AUC
Image only	96.58%	99.86%
Image + metadata	96.04%	99.85%

Metadata raised accuracy for seed 42 but lowered it for seeds 123 and 2026, so there is **no consistent benefit**. The link to Section 3.4: these three numbers describe how the image was acquired, not the patient. Had they helped, the model would have been learning acquisition style specific to this hospital, which would not be expected to hold at a rural camp with different equipment. Their failure to help is mildly reassuring, but it does not rule out that the image branch uses similar cues on its own.

8. Operating point, calibration and prevalence shift

8.1 Cost-sensitive threshold

A missed pneumonia costs 20 times a false alarm, and camp prevalence is taken as 90% NORMAL and 10% PNEUMONIA. In units of one false alarm, the expected cost per child is

C = 0.9 × FPR + 0.1 × 20 × FNR = 0.9 × FPR + 2 × FNR.

Thresholds were optimised per model across the three seeds (costs per 1,000 children):

Model	Mean cost at 0.5	Mean cost at optimised threshold	Reduction	Mean selected threshold
A0	65.60	**53.32**	18.7%	0.4517
A3	98.43	75.89	22.9%	0.0877

A0 has lower expected cost and higher specificity; A3 has marginally higher sensitivity. A0's threshold varied more across seeds, so it is sensitive to the validation split. A3's very low mean threshold (0.0877) shows it was pushed to flag aggressively to protect recall. These are **preliminary validation-derived operating points**.

8.2 Calibration

Platt scaling was fitted on a calibration subset and evaluated on a held-out subset for each seed.

Model	Brier before → after	Log loss before → after	ECE before → after
A0	0.01817 → 0.01686	0.06273 → 0.05852	0.02115 → 0.01602
A3	0.02936 → 0.02364	0.11523 → 0.08309	0.02983 → 0.01928

ROC-AUC and PR-AUC are unchanged, as expected from a monotonic transform. A0 is better calibrated than A3, but the evaluation subsets keep the dataset's high pneumonia share, so this says nothing about calibration at 10% prevalence.

8.3 Prior-probability (label-shift) correction

The source prevalence is about 74% pneumonia and the assumed camp prevalence is 10%. Calibrated probabilities were adjusted with prior-odds correction and scored by reweighting validation samples to 10% prevalence.

Model	Weighted Brier	Weighted log loss	Weighted ECE
A0	0.03033 → 0.00958	0.10695 → 0.03381	0.04994 → 0.01208
A3	0.04657 → 0.01235	0.16466 → 0.04811	0.08035 → 0.01421

This assumes the *only* difference between the dataset and the camp is class prevalence (label shift). That is almost certainly too optimistic, because equipment and population also differ. The assumption must be tested on the external hospital before any clinical interpretation.

9. What this means for the CEO's three instructions

1. "Prove what Rohit did wrong." Reasonable, and the audit provides evidence: a 16-image validation set (95% CI about 79–100% even at 100% accuracy), exact duplicate files, 26 unverified near-duplicate train-test candidates, no patient identifiers, and a 74%-pneumonia dataset on which predicting PNEUMONIA for everything already scores 74.33%.

2. "Rebuild it to 99% accuracy." This is the wrong target.

- At 9:1 NORMAL:PNEUMONIA prevalence, a model that never detects pneumonia scores **90% accuracy**, so accuracy hides exactly the failure that matters.
- Using HOG's validation results as an illustration (sensitivity 98.07%, specificity 95.52%): per 1,000 children at 10% prevalence the model catches about 98 of 100 sick children and falsely alarms on about 40 of 900 healthy ones, so only about **71% of positive calls would be correct**. Accuracy there would be about 95.8%, and it would still miss roughly 2 sick children per 100.
- The 16×16 shortcut model (NORMAL recall 63.43%) would falsely alarm on about 37% of healthy children, which is the same kind of failure the pilot saw.
- The right request is therefore a **minimum sensitivity with a bounded false-alarm rate at 10% prevalence**, plus verified performance on a different hospital, reported with confidence intervals and a cost-based operating point.

3. "Under 200 ms and within 25 MB." Reasonable. A0 has 1,941,665 parameters, about 7.8 MB in float32, so size is not the constraint. CPU latency has not been benchmarked yet.

10. Limitations and remaining work

- **Patient independence is unproven.** Random splits may place images of the same child in both train and validation. All internal accuracies may be optimistic.

- **Seeds differ in split**, so SDs combine split and training randomness.
- **The 26 near-duplicate pairs** need visual verification, and then the affected images should be removed or grouped.
- **The 16×16 shortcut source** is not isolated. Add a masked-lung or intensity-matched control.
- **Two 16×16 runs are reported** (89.08% and 91.76%). The repository should record which configuration and seed produced each, and the final report should use one.
- **Pending experiments:** input resolution and augmentation ablations (including naming the medically invalid augmentation, horizontal flip, which can swap the heart's side and so corrupt anatomical orientation), H4 GAP ablation, imbalance treatment vs threshold moving, 5-fold CV, tuning, ensembles, feature-space analysis, the once-only external-hospital test with a fix, `predict.py` and demo with out-of-distribution checks, and the CEO memo.
- **Latency** on a CPU-only laptop and the final TFLite or compressed model size remain to be measured.
- **Intended use:** this is a research prototype and not a clinically validated device.